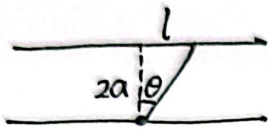
12.8 某根带电直线受到另一根带电直线的作用力一定垂直于该直线（对称性）

电场强度 $E=\int_{-\infty}^{+\infty}\frac{\lambda dl}{\left(\frac{2a}{\cos\theta}\right)^2\cdot 4\pi\varepsilon_0}\cdot\cos\theta$




$$=\int_{-\infty}^{+\infty}\frac{\lambda\cdot 2a\,dl}{4\pi\varepsilon_0(4a^2+l^2)^{\frac{3}{2}}}$$

$$=\int_{-\infty}^{+\infty}\frac{\lambda a\,dl}{2\pi\varepsilon_0(4a^2+l^2)^{\frac{3}{2}}}$$

$$=\frac{\lambda a}{2\pi\varepsilon_0}\cdot\frac{1}{2a^2}=\frac{\lambda}{4\pi\varepsilon_0 a}\Longleftarrow E=\frac{\lambda}{2\pi\varepsilon_0 r}$$，$r=2a$ 代入

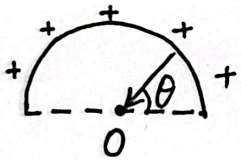
$$F=E\cdot\lambda=\frac{\lambda^2}{4\pi\varepsilon_0 a}$$

12.10 易知，该圆环上半部分带正电，下半部分带负电

且电荷分布关于y轴对称，电荷大小关于x轴对称

则圆心处的电场强度 $E=2E'$

（其中 $E'$ 为上半圆环在圆心处的电场强度）




由对称性可知，圆心处的电场强度沿y轴负方向

$$E'=\int_0^{\pi}\frac{\lambda\cdot R\,d\theta\cdot\sin\theta}{4\pi\varepsilon_0\cdot R^2}=\int_0^{\pi}\frac{\lambda_0}{4\pi\varepsilon_0 R}\sin\theta\,d\theta$$

$$=\frac{\lambda_0}{4\pi\varepsilon_0 R}\cdot\frac{\pi}{2}$$

$$=\frac{\lambda_0}{8\varepsilon_0 R}$$

则 $\vec{E}=2\vec{E'}$

$$=-\frac{\lambda_0}{4\varepsilon_0 R}\vec{j}$$

大小为 $\frac{\lambda_0}{4\varepsilon_0 R}$，方向沿y轴负方向

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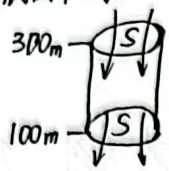
12.16 $\Delta\phi = \frac{\Delta q}{\varepsilon_0}$ 取底面积为S的封闭区间




$\Delta\phi = \oint_S E\,dS$

$= S(E_2 - E_1)$

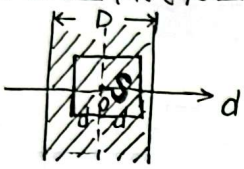
则 $\Delta q = \varepsilon_0 S(E_2 - E_1)$

平均体电荷密度 $\sigma = \frac{|\Delta q|}{e\Delta h S} = \frac{\varepsilon_0 \cdot |E_2 - E_1|}{e\Delta h}$

$= \frac{8.85\times10^{-12} \times |100-150|}{1.6\times10^{-19} \times 200}$

$= 1.38\times10^{7}\ (m^{-3})$

12.20 取过O且平行于厚壁区域，面积为S




由高斯定理有

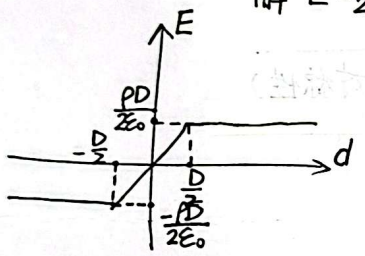
$ES = \frac{q}{\varepsilon_0}$

当 $0 \le d \le \frac{D}{2}$ 时，$q = 2\rho S d$

解 $E = \frac{\rho d}{\varepsilon_0}$

当 $\frac{D}{2} < d$ 时，$q = \rho S D$

解 $E = \frac{\rho D}{2\varepsilon_0}$




12.23

球内负电荷总量

$q_- = \int_0^\infty \rho(r)\,4\pi r^2 dr = \int_0^\infty -Ce^{-\frac{2r}{a_0}}\,4\pi r^2 dr$

$= -4\pi C\int_0^\infty r^2 e^{-\frac{2r}{a_0}} dr$

$= -4\pi C\int_0^{-\infty} \frac{a_0^2 t^2}{4} e^t \cdot -\frac{a_0}{2} dt$

$= \frac{\pi}{2} C a_0^3 \int_0^{-\infty} t^2 e^t dt$

$= \frac{\pi}{2} C a_0^3 \cdot (-2) = -C\pi a_0^3$

又 $q_- = -e$

则 $C = \frac{e}{\pi a_0^3}$

半径为$a_0$的球体内的负电荷总体

$Q_- = \int_0^{a_0} \rho(r)\,4\pi r^2 dr = \frac{e}{2}\int_0^{-a_0} t^2 e^t dt$

$= \frac{e}{2}(t^2 - 2t + 2)e^t \Big|_0^{-a_0}$

$= \frac{e}{2}\left[(a_0^2 + 2a_0 + 2)e^{-a_0} - 2\right]$

$= \frac{e}{2}(a_0^2 + 2a_0 + 2)e^{-a_0} - e$

净电荷 $Q = Q_- + Q_+$

$= \frac{e^{1-a_0}}{2}(a_0^2 + 2a_0 + 2)$

$= 1.08\times10^{-19}\,C$

距核$a_0$处的电场强度

$E = \frac{Q}{4\pi\varepsilon_0 a_0^2} = 3.46\times10^{11}\ V/m$

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# 作业纸

班级：　　姓名：　　学号：　　科目：　　第

12.26 铀核的体电荷密度

$$\rho = \frac{92e}{\frac{4}{3}\pi R^3} = \frac{92\times1.6\times10^{-19}}{\frac{4}{3}\pi\times(7.4\times10^{-15})^3} = 8.67\times10^{24}\ (C/m^3)$$

在τ子运动轨迹处的电场强度

$$E = \frac{\rho r}{3\varepsilon_0} = \frac{8.67\times10^{24}\times2.9\times10^{-15}}{3\times8.85\times10^{-12}} = 9.47\times10^{20}\ (V/m)$$

τ子受到的库仑力提供向心力

$$\frac{mv^2}{r} = eE \quad 解\ v = \sqrt{\frac{eEr}{m}} = 1.2\times10^7\ (m/s)$$

动能 $E_k = \frac{1}{2}mv^2 = \frac{1}{2}eEr = 2.2\times10^{-13}(J)$

角动量 $L = mvr = 1.1\times10^{-34}(kg\cdot m/s^2)$

频率 $\nu = \frac{v}{2\pi r} = 6.5\times10^{20}\ (Hz)$

12.27 氢原子负电荷体电荷密度

$$\rho = \frac{e}{\frac{4}{3}\pi r_0^3} = \frac{1.6\times10^{-19}}{\frac{4}{3}\pi\times(0.53\times10^{-10})^3} = 2.57\times10^{11}\ (C/m^3)$$

外加电场后，该设负电荷球心与质子距离为R

在质子处的电场

$$E_0 = \frac{\rho R}{3\varepsilon_0}$$

质子受电场力平衡则 $E - E_0 = 0$

$$解\ R = \frac{3\varepsilon_0 E}{\rho} = \frac{3\times8.85\times10^{-12}\times3\times10^6}{2.57\times10^{11}} = 3.1\times10^{-16}\ (m)$$

此时氢原子的"感生电偶极矩"

$$p = eR = 1.6\times10^{-19}\times3.1\times10^{-16} = 4.96\times10^{-35}\ (C\cdot m)$$

12.28 (1) 正立方体电通量

$$\phi = (E_2 - E_1)S = a^2\cdot(b\sqrt{a} - b\sqrt{a}) = a^2\sqrt{a}\cdot b(\sqrt{2}-1) = (\sqrt{2}-1)\times800\times(0.1)^{\frac{5}{2}}$$

$$= 1.05\ (N\cdot m^2\cdot C)$$

(2) 正立方体的总电荷

$$Q = \varepsilon_0\phi = 8.85\times10^{-12}\times1.05 = 9.29\times10^{-12}(C)$$

12.31 电偶极子在电场中受到的力矩

$$\vec{M} = \vec{P}\times\vec{E}$$

$$W = \int_{\frac{\pi}{2}}^{\theta} -\vec{M}\,d\theta = \int_{\frac{\pi}{2}}^{\theta} -p\cdot E\cdot\sin\theta\,d\theta = PE\cos\theta\Big|_{\frac{\pi}{2}}^{\theta} = PE\cos\theta$$

$$= \vec{P}\cdot\vec{E}$$

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